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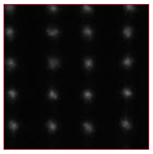

Speckle Imaging: PIXIS Helps Resolve Distant Stars

Elliott Horch from University of Massachusetts, Dartmouth uses speckle imaging to resolve distant binary stars, which otherwise can not be resolved due to atmosphere turbulence. The speckle imaging system consists of an optics box which tips and tilts th star image to a specific location on the CCD and a PIXIS: 2048B-2Kx2K, large format back illuminated CCD camera which is operated in a special "kinetics" mode of "step and expose" pattern with 50 ms exposures each location. The camera shutter remains open during the whole sequence.

In this way, stellar speckle patterns are captured which are not unlike the variation in brightness inside a laser spot when a laser pointer is reflected off a wall. Stellar speckle patterns allow for diffraction-limited image reconstruction, because each "speckle" in a short exposure image is a copy of the diffraction-limited point spread function. If you have, for example, a double star, you will get double speckles. This is a very useful technique for overcoming the blurring of images by the atmosphere.

PIXIS: 2048 provides several key advantages to astronomy applications:

- Low read noise (< 3 e-rms) and very high quantum efficiency (~95%)
- Large imaging area (27.6 mm x 27.6mm)
- High linearity (>99%) means reliable photometry
- Deep cooling for negligible dark noise

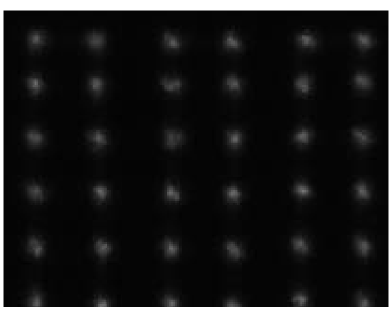

Figure 1. Speckle image captured by "tip-tilt" technique. Shown here is only approximately 1/4 the area of the PIXIS 2048B active area. Data courtesy of Elliott Horch, University of Massachusetts, Dartmouth, USA

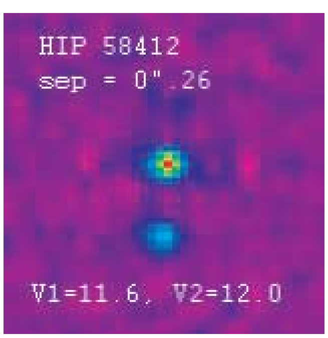

Figure 2. This is a diffraction-limited image reconstruction of a very faint star, more than 100 times fainter than can be seen by eye. It shows that two stars exist in the frame, only 0.26 arcseconds apart. (An arcsecond is the angle subtended by a dime at a distance of 2 miles.) If one took a normal image of the object, the atmosphere would blur things out so much that one would have no chance of seeing the second star.

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www.piacton.com
email: moreinfo@piacton.com
USA +1.877.4 PIACTON | Benelux +31 (347) 324989
France +33 (1) 60.86.03.65 | Germany +49 (0) 89.660.779.3
UK +44 (0) 28.38310171 | Asia/Pacific +65.6293.3130
China +86 135 0122 8135 | Japan +81.3.5639.2741